

# EECE7352 Computer Architecture

## Homework 1

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### Part E:

For this question I have taken, one General Purpose Benchmarking Suite (**SPEC**) and one Domain Specific Benchmarking Suite (**MLPerf**)

### SPEC Benchmark Suite

#### 1) What application domain or system architecture is the suite designed to evaluate?

The SPEC benchmark suite is primarily designed to evaluate the **performance of compute-intensive applications** on a wide range of hardware, from client PCs to servers. Its main goal is to provide a standardized, vendor-neutral measure of a system's performance.

- **Application Domain:** The suite targets **general-purpose computing**. Eg: SPEC CPU 2017 benchmark contains a set of real-world programs, including a C compiler, a chess program, a quantum computer simulator, and weather prediction models.
- **System Architecture:** It is designed to be architecture-agnostic, evaluating the performance of the entire system, including the **CPU, memory subsystem, and compiler technology**. It is widely used to compare processors from different vendors (e.g., Intel, AMD, ARM) and to assess the impact of different compilers and system configurations.

#### 2) How is performance evaluated, and what is the metric?

Performance in SPEC benchmarks is evaluated by measuring **how long a system takes** to run a set of reference programs compared to a base machine.

The results are then reported using two primary metrics:

- **SPEC speed** (Base result): This metric measures the **time it takes to complete a single task**.
- **SPEC rate** (Peak result): This metric measures the **throughput or capacity of the machine**.

#### 3) Cite 2 papers where the suite has been used in a research study.

**Paper 1:** (in APA format)

Bucek, J., Lange, K. D., & v. Kistowski, J. (2018, April). SPEC CPU2017: Next-generation compute benchmark. In *Companion of the 2018 ACM/SPEC International Conference on Performance Engineering* (pp. 41-42).

**Paper 2:** (APA)

Limaye, A., & Adegbiya, T. (2018, April). A workload characterization of the spec cpu2017 benchmark suite. In *2018 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)* (pp. 149-158). IEEE.

#### 4) Provide your thoughts on how the suite could be improved.

Although SPEC is an industry standard, it could be improved in several ways:

- **Increased Real-World Relevance:** The workloads, while based on real programs, can become old. A more frequent and agile process for incorporating newer, more relevant applications would enhance its relevance.
- **Inclusion of Security-Related Workloads:** Modern processors have dedicated hardware for security functions. Adding benchmarks that measure the performance impact of security mitigations (like those for Spectre and Meltdown) or the efficiency of cryptographic operations would reflect a critical aspect of modern computing.

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## MLPerf Benchmark Suite

### 1) What application domain or system architecture is the suite designed to evaluate?

The MLPerf benchmark suite is specifically designed to evaluate the performance of machine learning (ML) hardware, software, and services.

- **Application Domain:** It covers a wide range of common ML tasks, which are broken down into different suites such as Training, Inference and HPC. eg: the **training suite** measures the time it takes to train ML models to a target quality level. Tasks include image classification, object detection, natural language processing (NLP), and recommendation systems.
- **System Architecture:** MLPerf is used to benchmark a diverse set of systems, from massive multi-node GPU clusters used for **training large language models** down to tiny, low-power microcontrollers for on-device ML.

### 2) How is performance evaluated, and what is the metric?

MLPerf's evaluation methodology is tailored to the specific challenges of machine learning.

- **For Training:** The primary metric is **"time-to-accuracy."** It measures the time required to train a model on a specific dataset until it reaches a pre-defined quality target.
- **For Inference:** The metrics depend on the scenario:
  1. **Throughput:** For server or offline scenarios, the metric is the **number of inferences per second** (e.g., queries per second, or QPS).
  2. **Latency:** For real-time or single-stream scenarios (like in-car systems), the metric is the **time taken to complete a single inference**, often measured at a specific percentile (%).

### 3) Cite 2 papers where the suite has been used in a research study.

**Paper 1:** (in APA format)

Mattson, P., Cheng, C., Damos, G., Coleman, C., Micikevicius, P., Patterson, D., ... & Zaharia, M. (2020). MLperf training benchmark. *Proceedings of Machine Learning and Systems*, 2, 336-349.

**Paper 2:** (APA)

Banbury, C., Reddi, V. J., Torelli, P., Holleman, J., Jeffries, N., Kiraly, C., ... & Xuesong, X. (2021). MLperf tiny benchmark. *arXiv preprint arXiv:2106.07597*.

#### 4) Provide your thoughts on how the suite could be improved.

MLPerf has rapidly become the standard for ML performance, but there are areas for enhancement:

- **Cost and Power Efficiency Metrics:** While performance is crucial, **the total cost of ownership (TCO)** and **power efficiency** are equally important, especially for large-scale deployments. Officially incorporating metrics like '**performance-per-watt**' into the main results would provide a more practical evaluation.
- **Adaptability to Newer Models and Techniques:** The ML field evolves at an extremely fast pace. Models like Transformers have become dominant, and new techniques like **Mixture-of-Experts (MoE)** are emerging. MLPerf needs a very agile process to adopt and standardize benchmarks for these state-of-the-art models quickly. A lag could make the existing benchmarks less representative of cutting-edge workloads.